

Exercise 4: Order Items Join Optimization

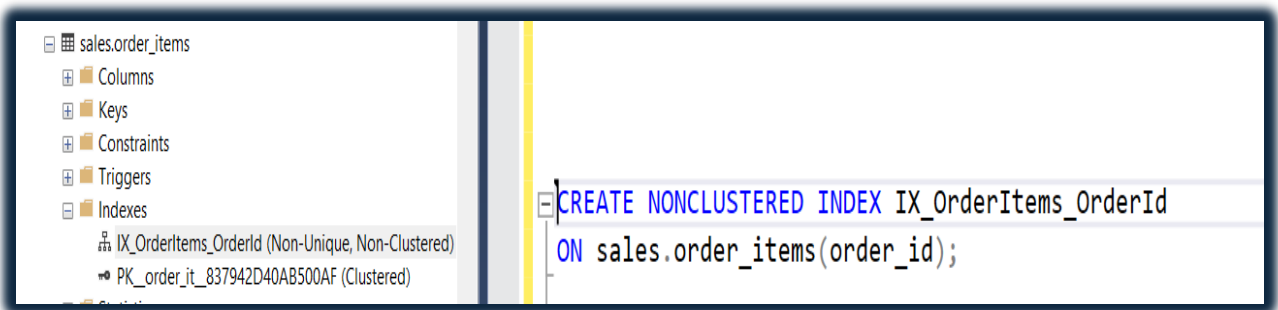
Sales reports require frequent joins between the orders table and the order_items table using order_id. These joins have become inefficient due to the growing data size.

- **A)** Identify a method to optimize these joins for better performance

To improve performance, we create a **Composite Covering Index** on the order_items table.

- **B)** Apply the optimization on the relevant table(s)

Create a composite non-clustered index that includes all columns used in the query:

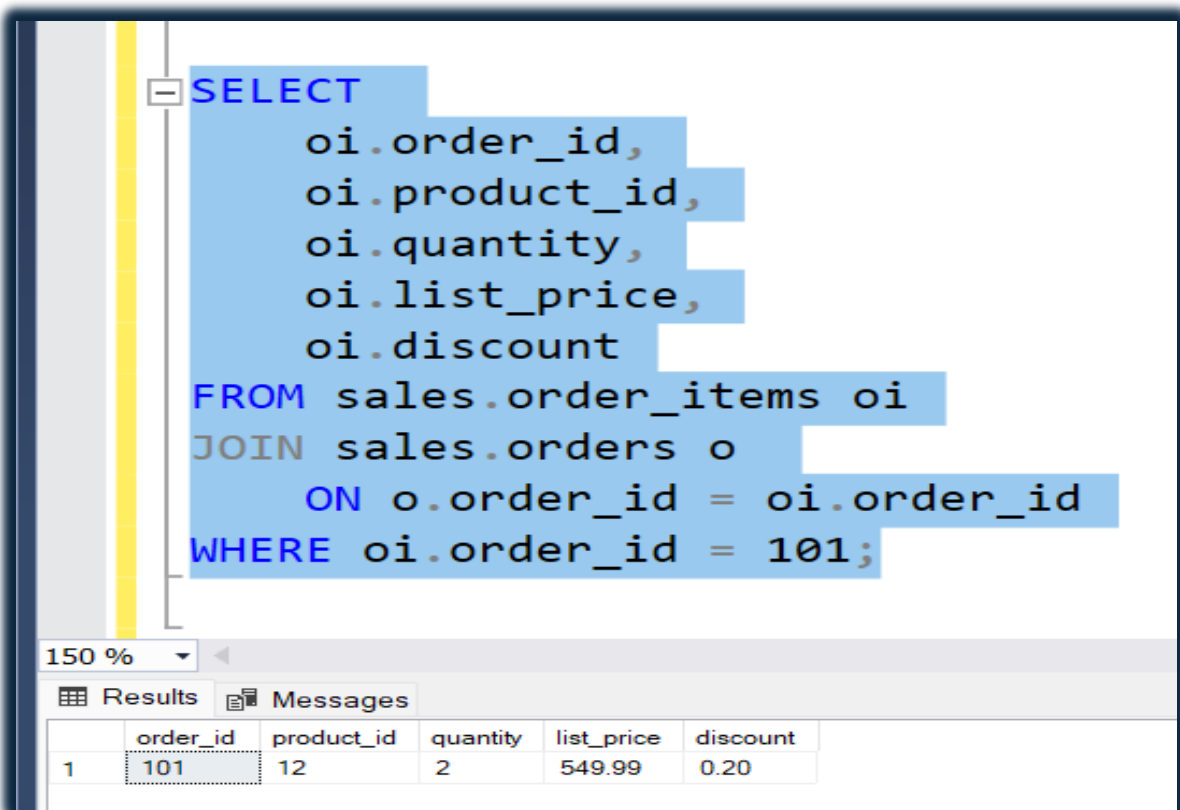


This index:

- Speeds up joins using order_id
 - Covers all columns required by the query
 - Reduces table lookups
-
- **C)** Retrieve the following details for a specific order:
 - order_id
 - product_id
 - quantity
 - list_price

- discount

Write a query to fetch order item details for a given order_id.



The screenshot shows a SQL IDE interface. The top pane contains a SQL query that selects order item details for a specific order ID. The query is as follows:

```
SELECT
    oi.order_id,
    oi.product_id,
    oi.quantity,
    oi.list_price,
    oi.discount
FROM sales.order_items oi
JOIN sales.orders o
    ON o.order_id = oi.order_id
WHERE oi.order_id = 101;
```

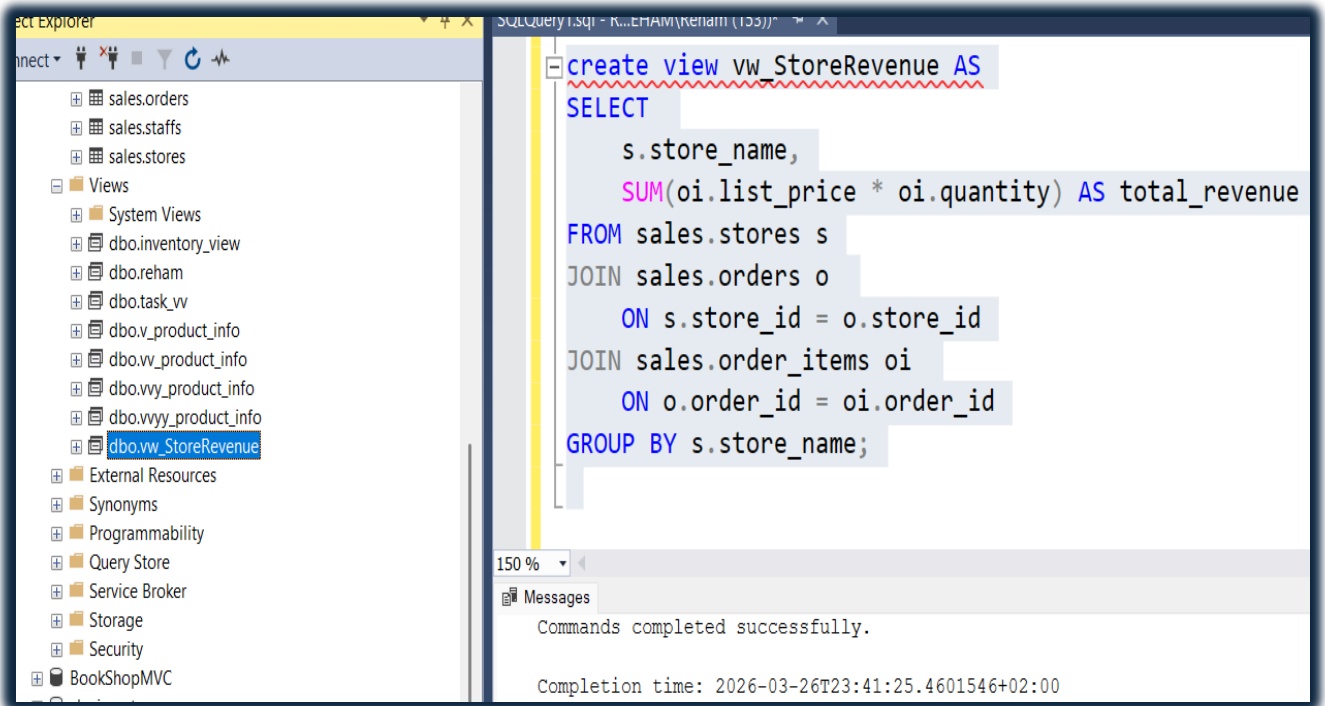
Below the query editor, there is a toolbar with a zoom level set to 150%. Two tabs are visible: 'Results' and 'Messages'. The 'Results' tab is active, displaying a table with the following data:

	order_id	product_id	quantity	list_price	discount
1	101	12	2	549.99	0.20

Exercise 5: Store Revenue View

The finance department needs a summary of total sales revenue per store.

- **A)** Implement a solution that allows quick revenue calculations (create view)



This view stores the revenue calculation logic and allows fast and simple retrieval later

- **B)** Retrieve the following details for each store:
 - store_name
 - total_revenue (sum of list_price * quantity)

Write a query to fetch total revenue per store.

```
= SELECT
    store_name,
    total_revenue
FROM vw_StoreRevenue;
```

150 %

Results Messages

	store_name	total_revenue
1	Santa Cruz Bikes	1790145.91
2	Rowlett Bikes	962600.76
3	Baldwin Bikes	5826242.21